

KEVIN GREIF (he/him)

ARTIFICIAL INTELLIGENCE — SCIENTIFIC RESEARCH — STATISTICAL METHODS

Physics Ph.D. with 5+ years of experience developing, deploying, and scaling ML for complex scientific problems. Published researcher (NeurIPS, SciPost Physics) in generative models (diffusion, flow-matching), transformer architectures, simulation-based inference, and large-scale distributed training. Seeking to accelerate science through applied AI, from research to production.

EXPERIENCE

Scientific AI methods development

2018 - Present

University of California, Irvine & University of Notre Dame

- Developed novel **latent diffusion and flow-matching models** for simulation-based inference in particle physics, learning to invert scientific simulations at state-of-the-art performance
- Designed generative architectures for **variable-size, heterogeneous data** (point clouds and sets), including equivariant networks encoding known physical symmetries
- Published research in peer-reviewed venues (**NeurIPS Main Conference 2023**, SciPost Physics 2024/2025)
- Presented research at international physics and machine learning conferences

Researcher at CERN - AI methods deployment

2020 - Present

ATLAS Collaboration

- Performed the highest-dimensional particle physics measurement in history, simultaneously characterizing ~ 150 correlated properties of each collision event. Previously intractable with conventional methods.
- Leveraged synthetic data to remove data-bound performance bottlenecks, making high-dimensional inference tractable
- Scaled **transformer** model training to 64 GPUs with PyTorch DDP on SLURM-managed HPC resources
- Performed hyperparameter optimization and rigorous uncertainty quantification for neural network classifiers on classification and simulation-based inference tasks
- Created and released open-source datasets, facilitating reproducibility and community adoption

Teaching, Leadership, and Service

- **ATLAS Early Career Scientist Board (2024-2026)**: Advocate for 1000+ early career scientists within international scientific collaboration, organize career development workshops and networking events
- **Teaching Assistant (2019-2021)**: Taught and mentored 100+ students in physics and computational methods courses

EDUCATION

University of California, Irvine

2020 - 2026

Ph.D. in Physics, Department of Physics and Astronomy — Advisor: Daniel Whiteson

Thesis defended May 2026; degree conferred September 2026

Focus: Artificial intelligence applications in particle physics research

University of Notre Dame

2016 - 2020

B.S. in Physics and Program of Liberal Studies, Magna Cum Laude

Honors: Outstanding Physics Major Award, Departmental award for outstanding writing, Phi Beta Kappa

SKILLS

AI & ML

- Transformers & attention models
- Diffusion & flow-matching
- Equivariant networks
- Simulation-based inference
- Distributed training (DDP)
- Hyperparameter optimization

Frameworks & Tools

- PyTorch & PyTorch Lightning
- TensorFlow, JAX
- Docker, Git, Weights & Biases
- SLURM / HPC clusters

Programming

- Python, C++
- CI/CD pipelines
- Testing frameworks

Data Science

- Large-scale data processing
- Statistical methods
- Uncertainty quantification
- Data visualization

Research & Collaboration

- Published author (NeurIPS, SciPost Physics)
- Peer review & technical writing
- 1000+ member intl. collaboration

SELECTED PUBLICATIONS

I only list publications in which I played a significant role. For a full list of publications, see my [Inspire HEP profile](#).

End-to-end latent variational diffusion models for inverse problems in high energy physics — [NeurIPS Main Conference](#)

Proposes a novel latent variational diffusion model applied to solving an important inverse problem in particle physics, demonstrating then state-of-the-art performance for generative model based approaches to this problem.

Full event particle-level unfolding with variable-length latent variational diffusion — [SciPost Physics](#)

Proposes a latent variational diffusion model for solving variable dimensional inverse problems in particle physics. Main development was the ability to process variable-length feature spaces.

Generative unfolding of jets and their substructure — [arXiv:2510.19906](#)

Proposes a flow-matching model for solving a very high (100+) and variable-dimensional inverse problem in particle physics.

Explicit or Implicit? Encoding Physics at the Precision Frontier — [arXiv:2603.08802](#)

Benchmarks symmetry-equivariant (explicit) vs. large-scale pretrained (implicit) approaches to encoding prior physics knowledge in neural networks.

Accuracy versus precision in boosted top tagging with the ATLAS detector — [JINST](#)

Benchmarking of various methods for an important classification problem in particle physics, with rigorous uncertainty quantification and release of a high-quality public dataset for the community.

The landscape of unfolding with machine learning — [SciPost Physics](#)

A community comparison study of unfolding methods in particle physics. I curated one of the comparison datasets used in this study.